Gaussian Elimination and Its Relation to Transform and Conquer

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1. What is Gaussian Elimination?

Gaussian Elimination is an algorithm for solving a system of n linear equations with n unknowns. It works by transforming the coefficient matrix into an **upper triangular matrix** using elementary row operations, then solving the system by **back substitution**.

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2. How is Gaussian Elimination related to Transform and Conquer?

- **Transform and Conquer** is a problem-solving technique that solves a problem by transforming it into a simpler or more convenient form and then solving that form.
- In Gaussian Elimination:
  - **Transform:** The original system (coefficient matrix) is transformed into an upper triangular matrix by eliminating entries below the main diagonal using row operations.
  - **Conquer:** Solve the transformed system easily using back substitution starting from the last equation.

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3. Main Steps of Gaussian Elimination

Step	Description
1. Matrix Representation	Write the system as an augmented matrix including coefficients and constants.
2. Forward Elimination	Use row operations to create zeros below the main diagonal, making the matrix upper triangular.
3. Back Substitution	Solve the system starting from the last equation moving upward.

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4. What is Pivoting?

- **Pivoting** is the process of swapping rows during forward elimination to avoid division by zero or very small numbers on the diagonal (pivot elements).
- It ensures numerical stability and avoids errors.

- Example: If the diagonal element is zero, swap with a lower row with a nonzero element in the same column.

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5. What is an Augmented Matrix?

An augmented matrix includes the coefficients of variables and the constants from the right side of the equations:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} & | & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & | & b_2 \\ \vdots & \vdots & \ddots & \vdots & | & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} & | & b_n \end{bmatrix}$$

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6. Practical Example

Solve the system:

$$\begin{cases} 2x + 3y - z = 5 \\ 4x + y + 2z = 6 \\ -2x + 5y + 3z = 7 \end{cases}$$

Step 1: Write the augmented matrix

$$\begin{bmatrix} 2 & 3 & -1 & | & 5 \\ 4 & 1 & 2 & | & 6 \\ -2 & 5 & 3 & | & 7 \end{bmatrix}$$

Step 2: Forward elimination — zero elements below first pivot (column 1)

- $R_2 \leftarrow R_2 - 2 \times R_1$
- $R_3 \leftarrow R_3 + 1 \times R_1$

New matrix:

$$\begin{bmatrix} 2 & 3 & -1 & | & 5 \\ 0 & -5 & 4 & | & -4 \\ 0 & 8 & 2 & | & 12 \end{bmatrix}$$

Step 3: Zero element below second pivot (column 2)

Calculate factor:

$$factor = \frac{8}{-5} = -\frac{8}{5}$$

Apply:

$$R_3 \leftarrow R_3 - factor \times R_2 = R_3 + \frac{8}{5} \times R_2$$

New matrix:

$$\begin{bmatrix} 2 & 3 & -1 & | & 5 \\ 0 & -5 & 4 & | & -4 \\ 0 & 0 & 8.4 & | & 5.6 \end{bmatrix}$$

Step 4: Back substitution

- From last row:

$$8.4z = 5.6 \implies z = \frac{5.6}{8.4} = \frac{2}{3} \approx 0.6667$$

- From second row:

$$\begin{gathered}-5y + 4z = -4 \\ -5y + 4 \times \frac{2}{3} = -4 \\ -5y + \frac{8}{3} = -4 \\ -5y = -4 - \frac{8}{3} = -\frac{20}{3} \\ y = \frac{4}{3} \approx 1.3333\end{gathered}$$

- From first row:

$$\begin{gathered}2x + 3y - z = 5 \\ 2x + 3 \times \frac{4}{3} - \frac{2}{3} = 5 \\ 2x + 4 - \frac{2}{3} = 5 \\ 2x = 5 - 4 + \frac{2}{3} = \frac{5}{3} \\ x = \frac{5}{6} \approx 0.8333\end{gathered}$$

7. Pseudocode for Gaussian Elimination with Pivoting

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Input: Augmented matrix A of size n x (n+1)

for i = 1 to n-1 do
    // Pivoting: find row with max absolute value in column i from
rows i to n
    maxRow = i
    for k = i+1 to n do
        if abs(A[k, i]) > abs(A[maxRow, i]) then
            maxRow = k
    if maxRow != i then
        swap rows A[i] and A[maxRow]

    // Forward elimination
    for j = i+1 to n do
        factor = A[j, i] / A[i, i]
        for k = i to n+1 do
            A[j, k] = A[j, k] - factor * A[i, k]

// Back substitution
x = array of size n
for i = n downto 1 do
    sum = 0
    for j = i+1 to n do
        sum = sum + A[i, j] * x[j]
    x[i] = (A[i, n+1] - sum) / A[i, i]

Output: solution vector x
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